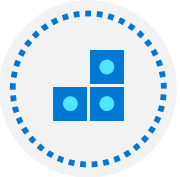


Training Course

Module 07: Azure Virtual Machines





Lesson 01: Configure Virtual Machines



Lesson 02: Configure Virtual Machine Availability

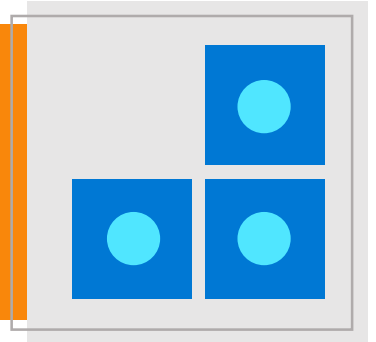


Lesson 03: Configure Virtual Machine Extensions



Lesson 04: Module 08 Lab

Lesson 01: Configure Virtual Machines



Review Cloud Services Responsibilities

Responsibility	SaaS	PaaS	IaaS	On-prem	
Information and data	Customer	Customer	Customer	Customer	RESPONSIBILITY ALWAYS RETAINED BY CUSTOMER
Devices (Mobile and PCs)	Customer	Customer	Customer	Customer	
Accounts and identities	Customer	Customer	Customer	Customer	
Identity and directory infrastructure	Microsoft	Customer	Customer	Customer	RESPONSIBILITY VARIES BY SERVICE TYPE
Applications	Microsoft	Customer	Customer	Customer	
Network controls	Microsoft	Customer	Customer	Customer	
Operating system	Microsoft	Microsoft	Customer	Customer	
Physical hosts	Microsoft	Microsoft	Microsoft	Customer	RESPONSIBILITY TRANSFERS TO CLOUD PROVIDER
Physical network	Microsoft	Microsoft	Microsoft	Customer	
Physical datacenter	Microsoft	Microsoft	Microsoft	Customer	

Microsoft Customer

Test and development, website hosting, storage, backup, recovery, high-performance computing, big data analysis, and extended data center

Plan Virtual Machines

Start with the network

Name the virtual machine

Choose a location

- Each region has different hardware and service capabilities
- Locate Virtual Machines as close as possible to your users and to ensure compliance and legal obligations

Consider pricing

- Compute costs
- Storage costs (consumption-based and reserved instances)



60+ Azure regions
Available in 140 countries

Determine Virtual Machine Sizing

A Series - Entry-level for dev/test

G Series – Memory and storage optimized

B Series – Economical bursting

H Series - High performance computing

D Series – General purpose compute

L Series – Storage optimized

Dc Series – Protect data in use

M Series – Memory optimized

E Series – In-memory hyper-threaded applications optimized

Mv2 Series – Largest memory optimized

F Series – Compute optimized

N Series – GPU enabled

Determine Virtual Machine Storage

Each Azure VM has two or more disks:

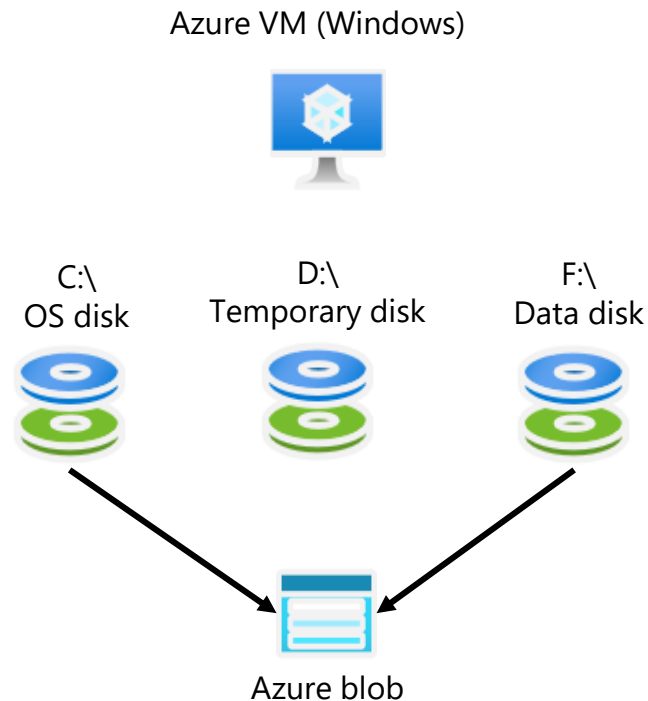
- OS disk
- Temporary disk (contents can be lost)
- Data disks (optional)

OS and data disks reside in Azure Storage accounts:

- Azure-based storage service
- Standard (HDD, SSD) or Premium (SSD), or Ultra (SSD)

When creating an Azure VM, you can choose between:

- Managed disks (recommended)
- Unmanaged disks



Create Virtual Machines in the Portal

Basic (required) – Project details, Administrator account, Inbound port rules

Disks – OS disk type, data disks

Networking – Virtual networks, load balancing

Management – Monitoring, Auto-shutdown, Backup

Advanced – Add additional configuration, agents, scripts or applications

Create a virtual machine

Basics

Disks

Networking

Management

Advanced

Tags

Review + create

Ubuntu Server 18.04 LTS

Ubuntu Server 18.04 LTS

Red Hat Enterprise Linux 7.7

SUSE Enterprise Linux 15 SP1

CentOS-based 7.7

Debian 10 "Buster" with backports kernel

Oracle Linux 7.7

Ubuntu Server 16.04 LTS

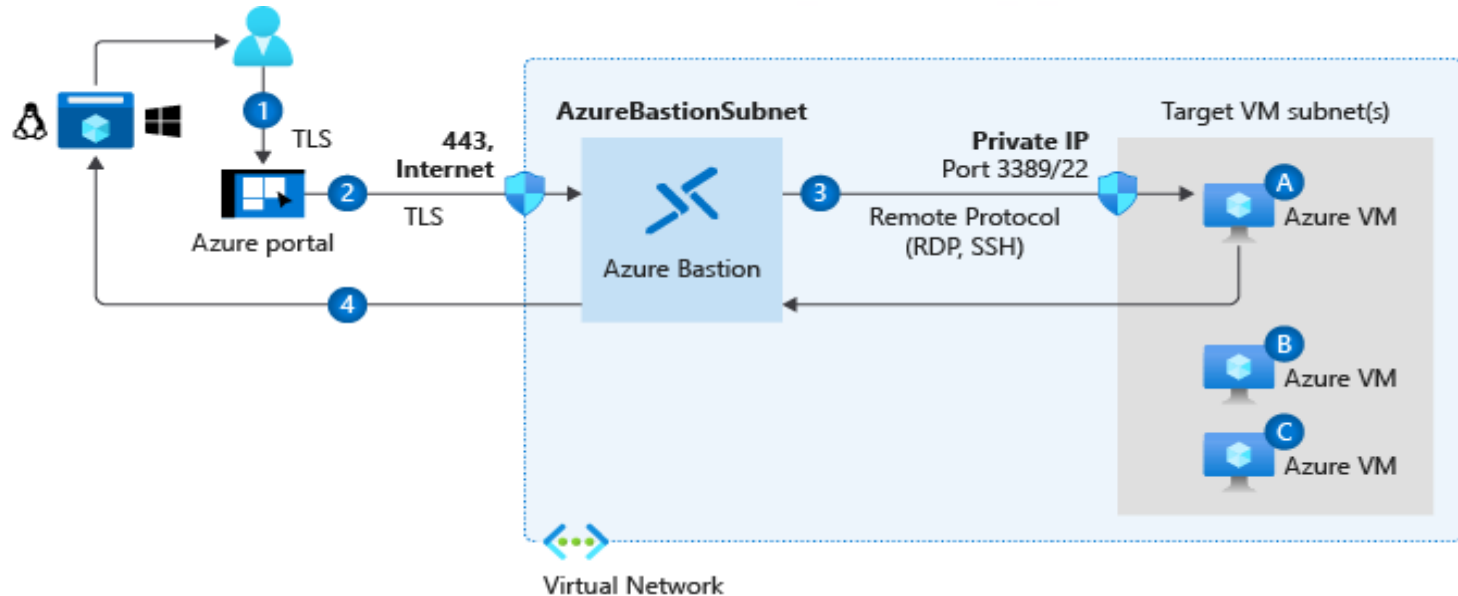
Windows Server 2019 Datacenter

Windows Server 2016 Datacenter

Windows Server 2012 R2 Datacenter

Windows 10 Pro, Version 1809

Connect to Virtual Machines



Bastion Subnet for RDP/SSH through the Portal over SSL

Remote Desktop Protocol for Windows-based Virtual Machines

Secure Shell Protocol for Linux based Virtual Machines

Connect to Windows Virtual Machines

Remote Desktop Protocol

(RDP) creates a GUI session and accepts inbound traffic on TCP port 3389

WinRM creates a command-line session so can run scripts



Connect to Linux Virtual Machines

Administrator account

Authentication type

Username * ⓘ

SSH public key * ⓘ

Provide an RSA public key in the single-line format (starting with "ssh-rsa") or the multi-line PEM format. You can generate SSH keys using ssh-keygen on Linux and OS X, or PuTTYGen on Windows.



[Learn more about creating and using SSH keys in Azure](#)

Authenticate with a
SSH public key or
password

SSH is an encrypted
connection protocol
that allows secure
logins over unsecured
connections

There are public and
private keys

Lesson 02: Configure Virtual Machine Availability



Unplanned Hardware Maintenance

When the platform predicts a failure, it will issue an **unplanned hardware maintenance** event

Action: Live migration

Unexpected Downtime

Unexpected Downtime is when a virtual machine fails unexpectedly

Action: Automatically migrate (heal)

Planned Maintenance

Planned Maintenance events are periodic updates made to the Azure platform

Action: No action

Setup Availability Sets

Instance details

Name * ⓘ

avset01



Region * ⓘ

(US) East US



Fault domains ⓘ

2

Update domains ⓘ

5

Use managed disks ⓘ

No (Classic) Yes (Aligned)

Configure multiple Virtual Machines in an Availability Set

Configure each application tier into separate Availability Sets

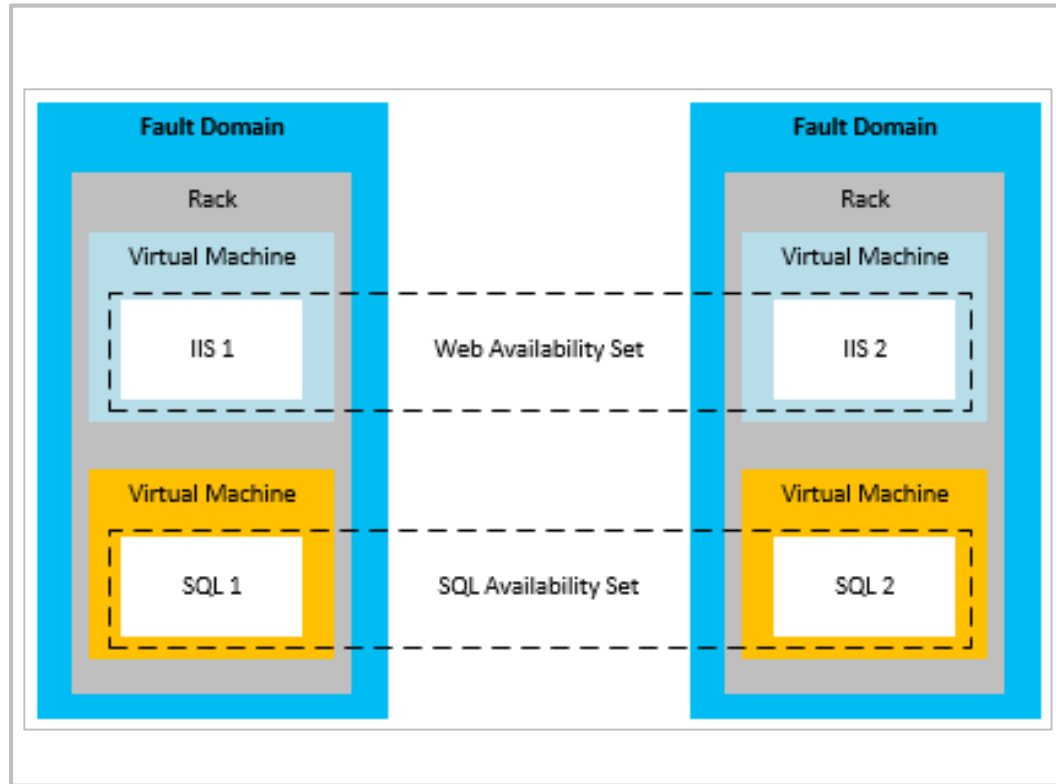
Combine a Load Balancer with Availability Sets

Use managed disks with the Virtual Machine

Review Update and Fault Domains

Update domains allows Azure to perform incremental or rolling upgrades across a deployment. During planned maintenance, only one update domain is rebooted at a time

Fault Domains are a group of Virtual Machines that share a common set of hardware, switches, that share a single point of failure. VMs in an availability set are placed in at least two fault domains



Review Availability Zones

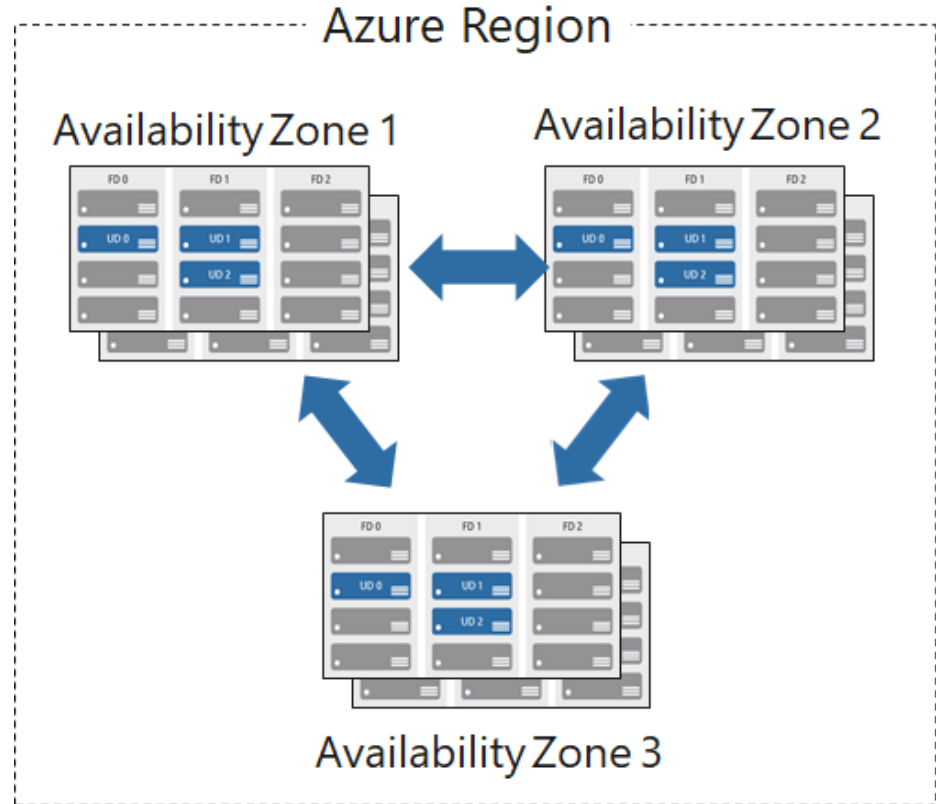
Unique physical locations in a region

Includes datacenters with independent power, cooling, and networking

Protects from datacenter failures

Combines update and fault domains

Provides 99.99% SLA

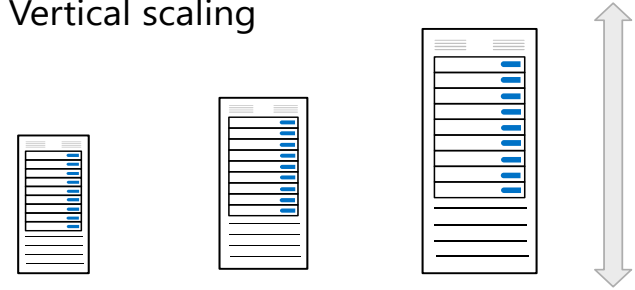


Compare Vertical to Horizontal Scaling

Vertical scaling (scale up and scale down) is the process of increasing or decreasing power to a single instance of a workload; usually manual

Horizontal scaling (scale out and scale in) is the process of increasing or decreasing the number of instances of a workload; frequently automated

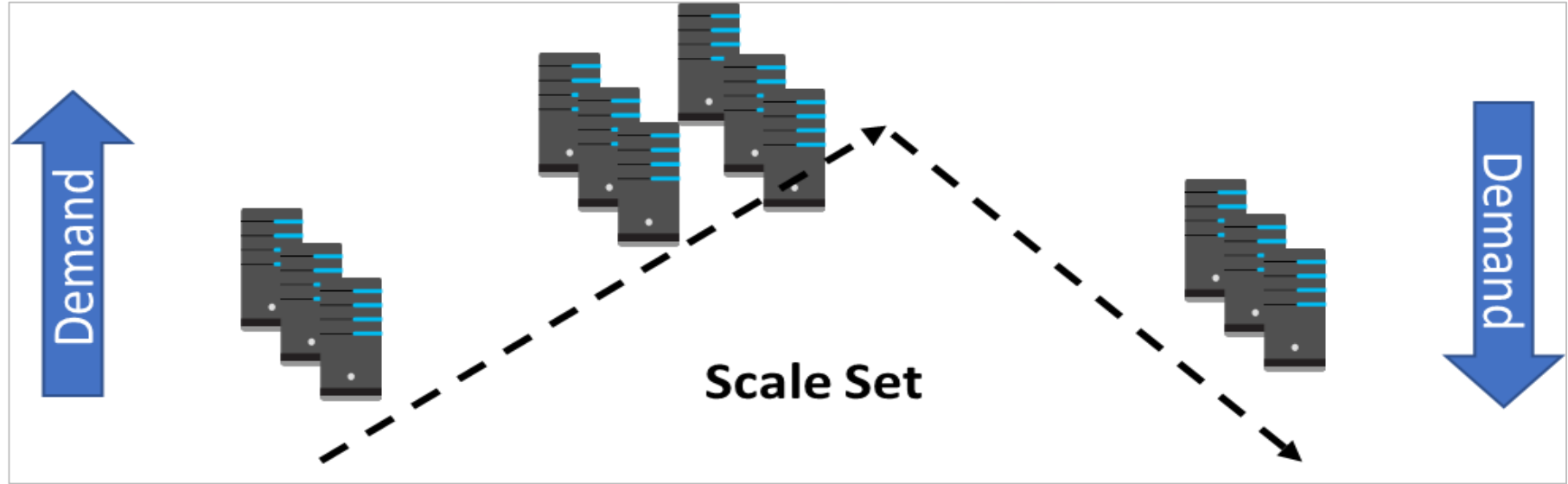
Vertical scaling



Horizontal scaling



Implement Scale Sets



Scale sets
deploy a set
of identical
VMs

No pre-
provisioning of
VMs is required

As demand
goes up
VMs are
added

As demand
goes down
VMs are
removed

The process can
be manual,
automated, or a
combination of
both

Create Scale Sets

Instance count. Number of VMs in the scale set (0 to 1000)

Instance size. The size of each virtual machine in the scale set

Azure Spot Instance. Unused capacity at a discounted rate

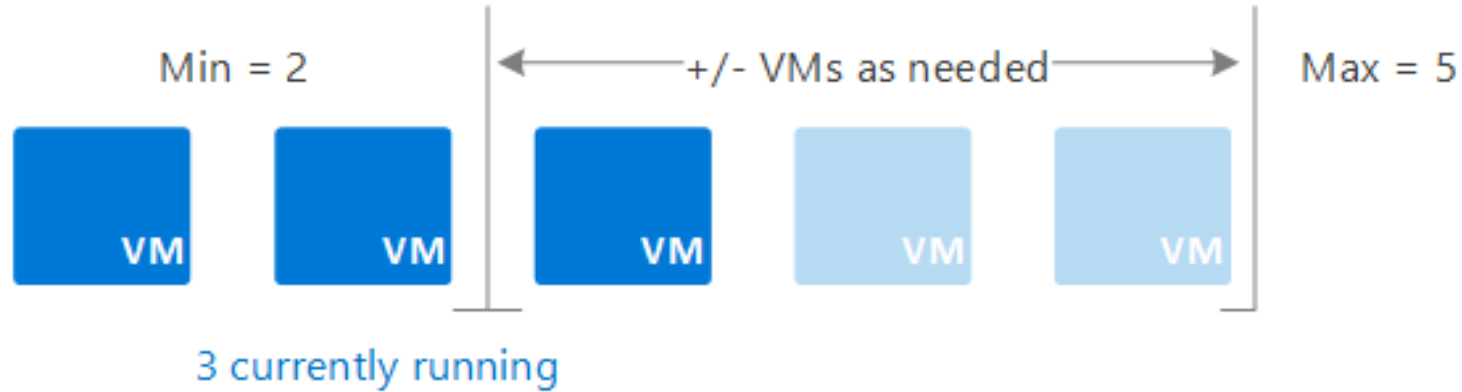
Use managed disks

Enable scaling beyond 100 instances

The screenshot shows the configuration page for an Azure Scale Set. The 'Instance' section includes the following settings:

- Initial instance count ***: A text input field containing the value '2'.
- Size ***: A dropdown menu showing 'Standard D2s v3' with a green border. Below it, the specifications '2 vcpus, 8 GiB memory (\$85.41/month)' and a 'Change size' link are visible.
- Azure Spot instance**: Radio buttons for 'Yes' and 'No', with 'No' selected.
- Use managed disks**: Radio buttons for 'No' and 'Yes', with 'Yes' selected.
- Allocation policy**: A section header for the scaling options.
- Enable scaling beyond 100 instances**: Radio buttons for 'No' and 'Yes', with 'No' selected.
- Spreading algorithm**: Radio buttons for 'Max spreading' and 'Fixed spreading (not recommended with zones)', with 'Fixed spreading' selected.

Implement Autoscale



Define rules to automatically adjust capacity

Scale out (increase) the number of VMs in the set

Scale in (reduce) the number of VMs in the set

Schedule events to increase or decrease at a fixed time

Reduces monitoring and optimizes performance

Configure Autoscale

Define a minimum, maximum, and default number of VM instances

Create more advanced scale sets with scale out and scale in parameters

Instance	
Initial instance count * ⓘ	2
Scaling	
Scaling policy ⓘ	☐ Manual ☒ Custom
Minimum number of VMs * ⓘ	1
Maximum number of VMs * ⓘ	10
Scale out	
CPU threshold (%) * ⓘ	75
Duration in minutes * ⓘ	10
Number of VMs to increase by * ⓘ	1
Scale in	
CPU threshold (%) * ⓘ	25
Number of VMs to decrease by * ⓘ	1

Thank you

